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CHAPTER 1

INTRODUCTION

1.1 Background

Skin cancer is one of the most common types of cancer worldwide. According to the World Health Organization (WHO), early diagnosis of skin cancer can significantly improve treatment outcomes and reduce mortality. Dermoscopic imaging is widely used by dermatologists to examine skin lesions. However, manual diagnosis requires extensive clinical expertise and is time-consuming. The increasing availability of medical image datasets and advances in Artificial Intelligence (AI) have made automated skin lesion classification an important research area.

Deep Learning, especially Convolutional Neural Networks (CNNs), has shown remarkable success in medical image analysis. Transfer learning models such as EfficientNet can achieve high classification performance while requiring less computational cost. However, most deep learning models behave as "black boxes," making it difficult for clinicians to understand how predictions are generated. Therefore, Explainable Artificial Intelligence (XAI) techniques such as Grad-CAM are becoming increasingly important for improving model transparency and trust.

1.2 Problem Statement

Accurate classification of different types of skin cancer remains a challenging task due to:

- High visual similarity among different lesion categories.
- Imbalanced class distribution in publicly available datasets.
- Lack of interpretability in conventional deep learning models.

- Dependence on experienced dermatologists for accurate diagnosis.

Therefore, there is a need for an explainable deep learning framework capable of accurately classifying multiple categories of skin lesions while providing visual explanations of its predictions.

1.3 Research Objectives

The main objective of this research is to develop an explainable deep learning framework for multi-class skin cancer classification.

The specific objectives are:

- To develop a CNN-based skin lesion classification model using EfficientNetB0.
- To classify dermoscopic images into seven skin lesion categories.
- To evaluate the model using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.
- To apply Grad-CAM for visual explanation of model predictions.

1.4 Research Questions

This study aims to answer the following questions:

1. Can EfficientNetB0 effectively classify seven categories of skin lesions?
2. How accurately can the proposed model classify dermoscopic images?
3. Can Grad-CAM improve the interpretability of the deep learning model?

1.5 Scope of the Study

This research focuses on:

- Multi-class skin lesion classification.
- HAM10000 dermoscopic image dataset.
- EfficientNetB0 transfer learning architecture.
- Explainable AI using Grad-CAM.
- Performance evaluation using standard classification metrics.

The study does not include clinical deployment or real-time diagnostic systems.

1.6 Significance of the Study

The proposed research contributes to the field of AI-assisted medical diagnosis by providing an explainable deep learning framework for skin cancer classification. The developed system may

assist dermatologists by improving diagnostic efficiency while increasing transparency through Grad-CAM visualization.

1.7 Organization of the Thesis

This thesis is organized into five chapters.

- **Chapter 1:** Introduction
- **Chapter 2:** Literature Review
- **Chapter 3:** Proposed Methodology
- **Chapter 4:** Experimental Results and Discussion
- **Chapter 5:** Conclusion and Future Work

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The rapid advancement of Artificial Intelligence (AI) and Deep Learning has significantly transformed the field of medical image analysis. In recent years, automated disease detection systems have become increasingly important due to their ability to improve diagnostic accuracy while reducing the workload of healthcare professionals. Skin cancer classification has emerged as one of the most active research areas because early diagnosis plays a vital role in successful treatment and patient survival.

Researchers have proposed numerous deep learning approaches for skin lesion classification using dermoscopic images. Traditional machine learning techniques required handcrafted feature extraction, whereas modern deep learning models automatically learn discriminative features from medical images. Furthermore, the integration of Explainable Artificial Intelligence (XAI) techniques has enhanced the interpretability of deep learning models, allowing clinicians to understand the reasoning behind model predictions.

This chapter reviews the existing literature related to skin cancer, deep learning, transfer learning, EfficientNetB0, Explainable AI, Grad-CAM, and previous research conducted in this field.

2.2 Skin Cancer

Skin cancer is one of the most common cancers worldwide and is caused by the abnormal growth of skin cells. It usually develops due to prolonged exposure to ultraviolet (UV) radiation from sunlight or artificial sources such as tanning devices. Early detection is crucial because timely treatment greatly increases the probability of successful recovery.

Skin cancer is generally categorized into melanoma and non-melanoma skin cancers. Melanoma is considered the most dangerous type because it spreads rapidly to other parts of the body if left untreated. Non-melanoma cancers include Basal Cell Carcinoma (BCC), Squamous Cell Carcinoma (SCC), and several benign skin lesions.

The HAM10000 dataset, widely used in medical image classification research, contains seven categories of dermoscopic skin lesion images:

- Actinic Keratoses (akiec)
- Basal Cell Carcinoma (bcc)
- Benign Keratosis (bkl)
- Dermatofibroma (df)
- Melanoma (mel)
- Melanocytic Nevus (nv)
- Vascular Lesions (vasc)

These categories provide a diverse benchmark for evaluating deep learning models in multi-class skin lesion classification.

2.3 Deep Learning in Medical Image Classification

Deep Learning is a branch of Artificial Intelligence that enables computers to learn complex patterns directly from data. Convolutional Neural Networks (CNNs) have become the dominant deep learning architecture for medical image analysis due to their excellent ability to extract spatial features automatically.

CNNs consist of multiple convolutional layers, pooling layers, activation functions, and fully connected layers. These components work together to identify hierarchical image features ranging from simple edges to complex anatomical structures.

Several studies have demonstrated that CNN-based models outperform traditional machine learning algorithms in medical image classification tasks. Applications include skin cancer detection, brain tumor classification, diabetic retinopathy diagnosis, pneumonia detection, and breast cancer analysis.

The major advantages of deep learning include:

- Automatic feature extraction
- High classification accuracy
- Scalability for large datasets
- Robust performance on complex medical images

However, deep learning models are often criticized for their lack of interpretability, making Explainable AI techniques increasingly important.

2.4 Transfer Learning

Transfer Learning is a deep learning technique in which a model trained on a large dataset is reused for a different but related task. Instead of training a neural network from scratch, a pre-trained model is fine-tuned using a new dataset. This approach significantly reduces training time, computational cost, and the amount of labeled data required.

In medical image classification, transfer learning has become one of the most widely used techniques because medical datasets are often limited in size. Models pre-trained on ImageNet have already learned rich visual features such as edges, textures, and shapes, which can be effectively adapted to medical imaging tasks.

Several popular pre-trained models include VGG16, ResNet50, InceptionV3, MobileNet, DenseNet, EfficientNet, and Vision Transformer (ViT). Among these models, EfficientNet has gained significant attention due to its excellent balance between classification accuracy and computational efficiency.

The advantages of transfer learning include:

- Reduced training time.
- Improved model accuracy.

- Better generalization performance.
- Lower computational requirements.
- Effective performance on small medical datasets.

For these reasons, transfer learning has become a standard practice in modern medical image analysis.

2.5 EfficientNetB0

EfficientNet is a family of Convolutional Neural Networks proposed by Google Research. Unlike conventional CNN architectures that increase network depth or width independently, EfficientNet introduces a compound scaling method that uniformly scales the network's depth, width, and input resolution.

EfficientNetB0 is the baseline model of the EfficientNet family. Despite having fewer parameters than many traditional CNN models, it achieves competitive classification performance while maintaining computational efficiency.

The EfficientNetB0 architecture consists of several key components:

- Convolutional Layers
- Batch Normalization
- Swish Activation Function
- Mobile Inverted Bottleneck Convolution (MBConv)
- Squeeze-and-Excitation (SE) Blocks
- Global Average Pooling
- Fully Connected Classification Layer

The MBConv block improves feature extraction while reducing computational complexity. The Squeeze-and-Excitation block enables the network to emphasize important feature channels by assigning adaptive weights.

Due to its lightweight architecture and strong feature learning capability, EfficientNetB0 has become a popular choice for medical image classification tasks, including skin lesion analysis.

In this research, EfficientNetB0 was selected as the backbone network because it provides a good balance between accuracy, computational efficiency, and model size.

2.6 Explainable Artificial Intelligence (XAI)

Although deep learning models achieve high classification accuracy, they often operate as "black-box" systems, making it difficult to understand how predictions are generated. In healthcare applications, model interpretability is essential because medical professionals require trustworthy explanations before accepting AI-assisted decisions.

Explainable Artificial Intelligence (XAI) refers to a collection of techniques that make machine learning models more transparent and understandable. Instead of providing only a predicted class label, XAI methods highlight the features or image regions that influenced the model's decision.

The main objectives of XAI include:

- Improving model transparency.
- Increasing clinician trust.
- Supporting medical decision-making.
- Identifying incorrect model predictions.
- Enhancing reliability in healthcare applications.

Several XAI techniques have been proposed for image classification, including:

- Grad-CAM
- Grad-CAM++
- LIME
- SHAP
- Integrated Gradients

Among these methods, Grad-CAM is one of the most widely adopted techniques for CNN-based medical image analysis because it produces intuitive visual heatmaps without modifying the original network architecture.

2.7 Gradient-weighted Class Activation Mapping (Grad-CAM)

Gradient-weighted Class Activation Mapping (Grad-CAM) is an Explainable AI technique that visualizes the regions of an image responsible for a deep learning model's prediction. It generates a heatmap using the gradients flowing into the final convolutional layer of a CNN.

Instead of merely predicting the class label, Grad-CAM highlights the most influential areas in the input image. Regions with higher importance appear in warmer colors (such as red and yellow), while less important regions appear in cooler colors (such as blue).

The Grad-CAM process consists of the following steps:

1. Pass the input image through the trained CNN.
2. Compute the predicted class score.
3. Calculate the gradients of the predicted class with respect to the final convolutional layer.
4. Average the gradients to obtain channel importance weights.
5. Generate a weighted combination of feature maps.
6. Apply the ReLU activation function.
7. Resize the resulting heatmap to match the original image dimensions.

8. Overlay the heatmap on the original dermoscopic image.

The advantages of Grad-CAM include:

- Easy interpretation of model predictions.
- No modification of the trained network.
- Compatibility with most CNN architectures.
- Improved transparency in medical diagnosis.

In this research, Grad-CAM was employed to explain the predictions of the EfficientNetB0 model. The generated heatmaps allow visualization of the lesion regions that contributed most to the classification results, thereby improving the interpretability and trustworthiness of the proposed framework.

2.8 Related Works

Several researchers have applied deep learning techniques for automated skin cancer classification using dermoscopic images. Their findings demonstrate the effectiveness of convolutional neural networks and transfer learning in improving classification performance.

Esteva et al. (2017) developed a deep convolutional neural network for skin cancer classification using a large dataset of clinical images. Their model achieved dermatologist-level performance in classifying skin lesions, demonstrating the potential of deep learning in medical diagnosis.

Tschandl et al. (2018) introduced the HAM10000 dataset, which contains over ten thousand dermoscopic images belonging to seven different skin lesion categories. The dataset has become one of the most widely used benchmarks for multi-class skin lesion classification.

Tan and Le (2019) proposed the EfficientNet family of convolutional neural networks. Their compound scaling approach significantly improved image classification accuracy while reducing computational complexity. EfficientNet models have since been widely adopted in medical image analysis.

Selvaraju et al. (2017) proposed Gradient-weighted Class Activation Mapping (Grad-CAM), an Explainable Artificial Intelligence technique that highlights important image regions responsible for CNN predictions. Grad-CAM has become one of the most popular visualization methods for interpreting deep learning models.

Several recent studies have combined EfficientNet with Explainable AI techniques to improve the transparency of medical image classification systems. Although these methods have demonstrated promising performance, many studies primarily focus on binary classification or do not provide sufficient visual explanations for multi-class skin lesion diagnosis.

The literature indicates that combining EfficientNetB0 with Grad-CAM provides an effective balance between classification accuracy and model interpretability, making it suitable for medical diagnostic applications.

2.9 Research Gap

Although numerous studies have investigated automated skin cancer classification, several research gaps remain.

First, many existing studies focus mainly on binary classification instead of classifying multiple categories of skin lesions. Multi-class classification is more challenging because several lesion categories exhibit highly similar visual characteristics.

Second, many deep learning models provide only prediction results without explaining the reasoning behind their decisions. Such "black-box" behavior limits the practical adoption of AI systems in healthcare, where clinicians require interpretable and trustworthy predictions.

Third, class imbalance remains a major challenge in publicly available skin lesion datasets such as HAM10000. Minority classes often receive lower classification accuracy than majority classes, reducing the overall robustness of the model.

Finally, although EfficientNet-based models have demonstrated excellent classification performance, relatively few studies combine EfficientNetB0 with Explainable Artificial Intelligence techniques such as Grad-CAM for multi-class skin lesion classification.

To address these limitations, this research proposes an explainable deep learning framework that combines EfficientNetB0 with Grad-CAM for seven-class skin cancer classification using the HAM10000 dataset.

2.10 Chapter Summary

This chapter reviewed the existing literature related to skin cancer classification using deep learning techniques. It discussed the characteristics of skin cancer, the application of Convolutional Neural Networks in medical image analysis, the advantages of transfer learning, the EfficientNetB0 architecture, and Explainable Artificial Intelligence techniques such as Grad-CAM.

Furthermore, previous research on automated skin lesion classification was reviewed to identify current trends and limitations. The identified research gaps justify the need for an explainable multi-class skin cancer classification framework. The next chapter presents the proposed methodology, including dataset preparation, preprocessing techniques, model architecture, training strategy, evaluation metrics, and Grad-CAM implementation.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Introduction

This chapter presents the methodology adopted for developing the proposed explainable deep learning framework for multi-class skin cancer classification. The complete workflow includes dataset collection, image preprocessing, data augmentation, transfer learning using EfficientNetB0, model training, performance evaluation, and visual explanation using Grad-CAM.

The overall objective of the proposed methodology is to develop an accurate and interpretable deep learning model capable of classifying seven different categories of skin lesions using dermoscopic images.

3.2 Proposed Framework

The proposed framework consists of several sequential stages beginning from dataset preparation and ending with model interpretation. Figure 3.1 illustrates the overall workflow of the proposed system.

The framework includes the following steps:

1. Collection of the HAM10000 dataset.
2. Image preprocessing and resizing.
3. Data augmentation.
4. Data splitting into training, validation, and testing sets.
5. Feature extraction using EfficientNetB0.
6. Fine-tuning of the classification model.
7. Performance evaluation.
8. Explainability using Grad-CAM.

The overall workflow ensures both high classification performance and improved interpretability of model predictions.

3.3 Dataset Description

The HAM10000 (Human Against Machine with 10000 Training Images) dataset was used in this research. It is one of the most widely used public datasets for dermoscopic skin lesion classification.

The dataset contains **10,015 dermoscopic images** belonging to seven different skin lesion categories. Each image is associated with a diagnostic label representing one of the skin lesion classes.

The seven categories included in the dataset are:

Class	Description
akiec	Actinic Keratoses
bcc	Basal Cell Carcinoma
bkl	Benign Keratosis
df	Dermatofibroma
mel	Melanoma
nv	Melanocytic Nevus
vasc	Vascular Lesions

The dataset was collected from different clinical sources and contains lesions captured under various imaging conditions, making it suitable for evaluating deep learning models.

3.4 Data Preprocessing

Before training the deep learning model, several preprocessing steps were performed to improve data quality and ensure consistency.

The preprocessing pipeline included:

- Image loading
- Image resizing
- Pixel normalization
- Label encoding
- Dataset splitting

All dermoscopic images were resized to **224 × 224 pixels**, which matches the required input size of EfficientNetB0.

Pixel values were normalized to the range **0–1** by dividing each pixel intensity by 255.

Categorical class labels were converted into numerical labels for model training.

The dataset was divided into three subsets:

Dataset	Percentage
Training	80%
Validation	10%

Dataset	Percentage
Testing	10%

This split ensured unbiased model evaluation while preventing overfitting.

3.5 Data Augmentation

Data augmentation was applied to improve the generalization capability of the deep learning model.

Several image transformation techniques were employed during training, including:

- Random rotation
- Horizontal flipping
- Zooming
- Width shifting
- Height shifting
- Shearing

These augmentation techniques artificially increased the diversity of the training dataset without collecting additional medical images.

Data augmentation also reduced the risk of overfitting by exposing the model to multiple variations of the same lesion image.

3.6 Proposed Model Architecture

The proposed classification model is based on **EfficientNetB0**, a state-of-the-art convolutional neural network architecture that provides high classification accuracy while maintaining computational efficiency. EfficientNetB0 was selected as the backbone network due to its balanced scaling strategy and relatively low number of trainable parameters.

A transfer learning approach was adopted by loading EfficientNetB0 with pre-trained ImageNet weights. The original classification layer was removed and replaced with custom fully connected layers suitable for seven-class skin lesion classification.

The final architecture consists of the following components:

- Input Layer ($224 \times 224 \times 3$)
- EfficientNetB0 Feature Extractor
- Global Average Pooling Layer
- Dense Layer
- Dropout Layer
- Output Layer with Softmax Activation

The Softmax activation function produces probability scores for each of the seven skin lesion categories, and the class with the highest probability is selected as the final prediction.

3.7 Model Training

The proposed model was trained using the TensorFlow and Keras deep learning frameworks. Transfer learning was applied to utilize the pre-trained knowledge of EfficientNetB0 while fine-tuning the model for skin lesion classification.

The model was compiled using the **Adam optimizer** with a learning rate of **0.0001**. The **Sparse Categorical Crossentropy** loss function was selected because the dataset labels were encoded as integer values.

To improve training performance and reduce overfitting, several callback functions were employed:

- EarlyStopping
- ReduceLROnPlateau
- ModelCheckpoint

EarlyStopping monitors the validation loss and stops training if no improvement is observed for several consecutive epochs. ReduceLROnPlateau automatically decreases the learning rate whenever the validation performance stagnates. ModelCheckpoint stores the best-performing model during training.

The model was trained for **10 epochs** using the training dataset, while validation data were used to monitor generalization performance.

3.8 Evaluation Metrics

To evaluate the performance of the proposed model, several standard classification metrics were employed.

Accuracy

Accuracy measures the proportion of correctly classified images among the total number of images.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

Precision measures how many predicted positive samples are actually positive.

Precision=TP+FPTP

Recall

Recall measures how many actual positive samples are correctly identified.

Recall=TP+FNTTP

F1-score

The F1-score is the harmonic mean of Precision and Recall.

$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

Confusion Matrix

The Confusion Matrix provides a detailed representation of classification performance by comparing predicted labels with true labels for all seven skin lesion categories.

3.9 Grad-CAM Implementation

To improve model interpretability, **Gradient-weighted Class Activation Mapping (Grad-CAM)** was integrated into the proposed framework.

Grad-CAM visualizes the image regions that contribute most significantly to the model's prediction. Instead of treating the deep learning model as a black box, the generated heatmaps provide intuitive explanations that highlight the lesion regions influencing the classification decision.

The Grad-CAM implementation consists of the following steps:

1. Load the trained EfficientNetB0 model.
2. Pass the input dermoscopic image through the network.
3. Compute gradients with respect to the last convolutional layer.
4. Calculate feature importance weights.
5. Generate the Grad-CAM heatmap.
6. Resize the heatmap to the original image size.
7. Overlay the heatmap on the original dermoscopic image.

The resulting visualization assists clinicians in understanding why the model predicts a particular skin lesion category.

3.10 Experimental Setup

All experiments were conducted using the **Kaggle Notebook** environment with GPU acceleration.

The software and hardware configuration used in this research is summarized below.

Component	Specification
Programming Language	Python
Deep Learning Framework	TensorFlow / Keras
Development Platform	Kaggle Notebook
GPU	NVIDIA Tesla T4
Dataset	HAM10000
Image Size	224×224
Model	EfficientNetB0
Optimizer	Adam
Learning Rate	0.0001
Epochs	10
Evaluation Metrics	Accuracy, Precision, Recall, F1-score, Confusion Matrix

3.11 Chapter Summary

This chapter described the methodology used to develop the proposed explainable deep learning framework for multi-class skin cancer classification. It discussed the HAM10000 dataset, preprocessing pipeline, data augmentation techniques, EfficientNetB0 architecture, model training strategy, evaluation metrics, Grad-CAM implementation, and experimental setup.

The proposed methodology combines transfer learning with Explainable Artificial Intelligence to produce an accurate and interpretable skin lesion classification system. The following chapter presents the experimental results, performance evaluation, and discussion of the proposed framework.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the experimental results obtained from the proposed explainable deep learning framework for multi-class skin cancer classification. The performance of the EfficientNetB0 model is evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrix. Furthermore, Grad-CAM visualizations are presented to explain the prediction behavior of the trained model.

The experiments were conducted using the HAM10000 dataset, and the results demonstrate the effectiveness of the proposed framework in classifying seven categories of skin lesions.

4.2 Model Training Performance

The EfficientNetB0 model was trained for ten epochs using the training dataset. During training, both the training accuracy and validation accuracy improved steadily, indicating that the model successfully learned discriminative features from dermoscopic images.

Similarly, the training loss and validation loss gradually decreased, showing stable convergence without severe overfitting. The use of transfer learning, data augmentation, EarlyStopping, and learning rate reduction contributed to improved generalization performance.

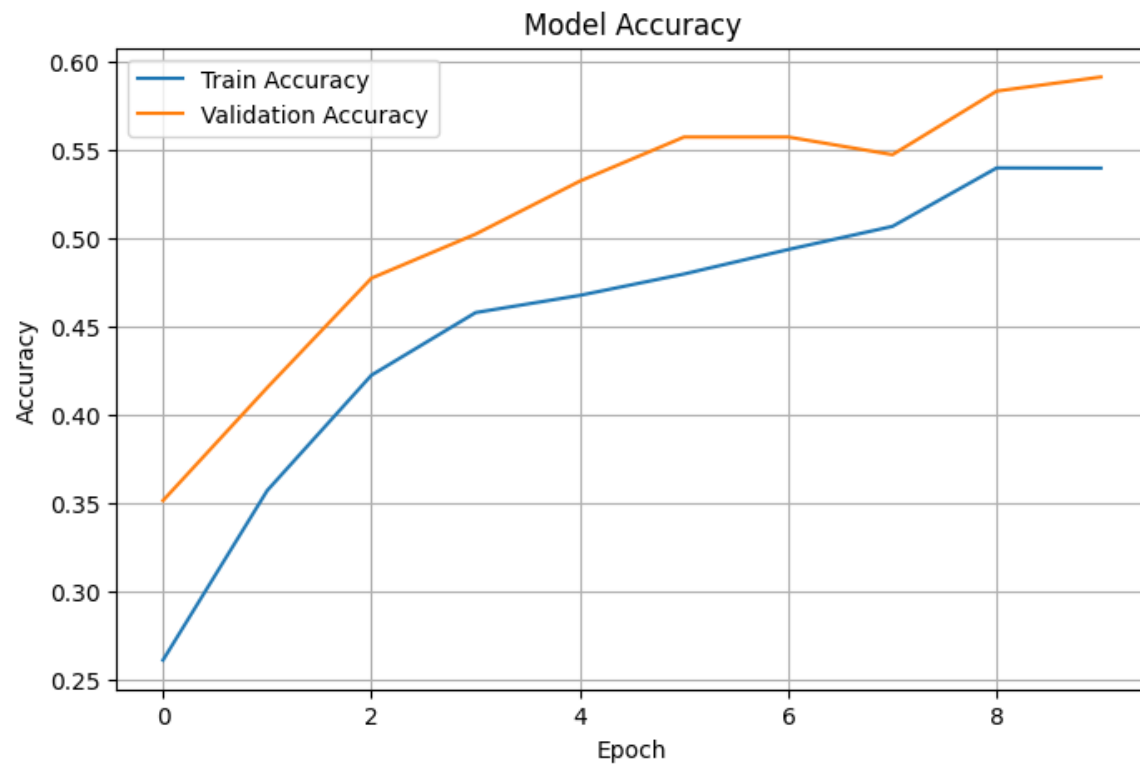


Figure 4.1: Training Accuracy Curve

The training accuracy increased consistently across epochs, demonstrating effective learning throughout the training process.

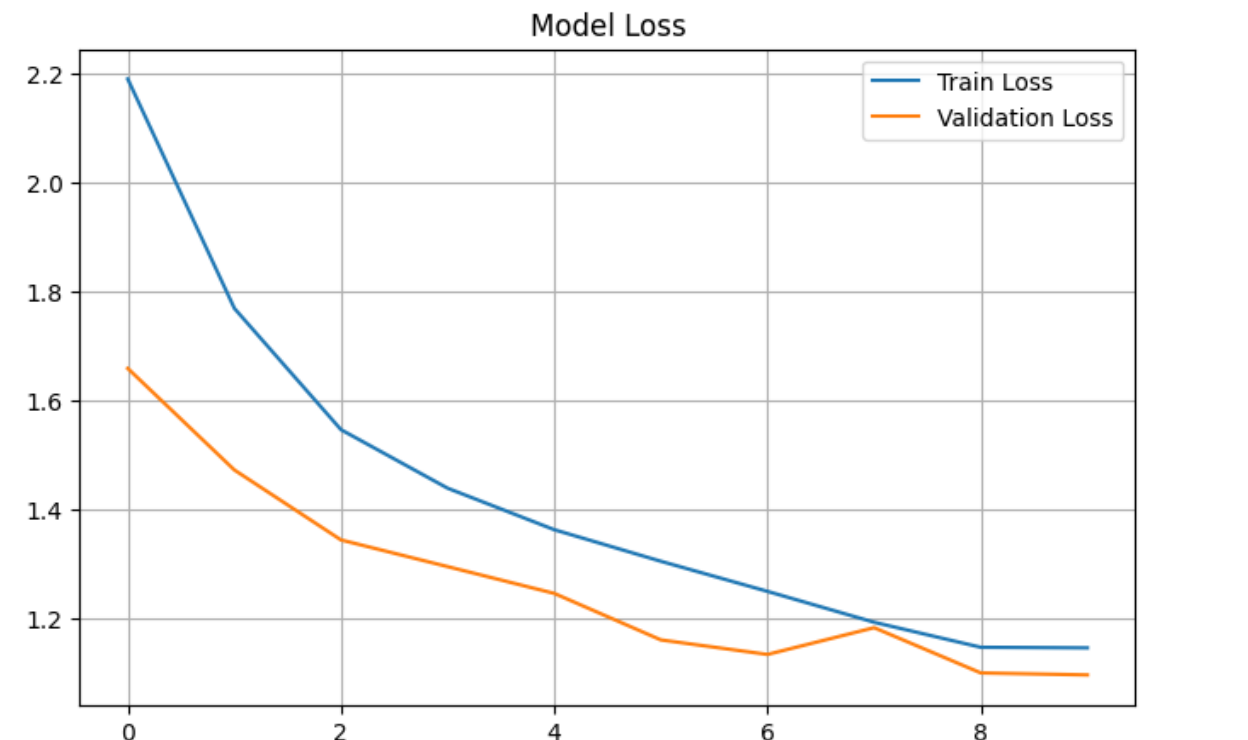


Figure 4.2: Training Loss Curve

The training and validation loss decreased progressively, indicating successful optimization of the proposed model.

4.3 Test Performance

After training, the model was evaluated using the unseen testing dataset.

The proposed EfficientNetB0 model achieved an overall **testing accuracy of 59.38%**, demonstrating satisfactory classification performance for seven-class skin lesion recognition.

Table 4.1 Test Performance

Metric	Value
Test Accuracy	59.38%

Although the overall accuracy is moderate, the model successfully classified the majority of Melanocytic Nevus (nv) images while maintaining reasonable performance across the remaining skin lesion categories.

4.4 Confusion Matrix Analysis

The confusion matrix provides a detailed visualization of the classification performance for each skin lesion category.

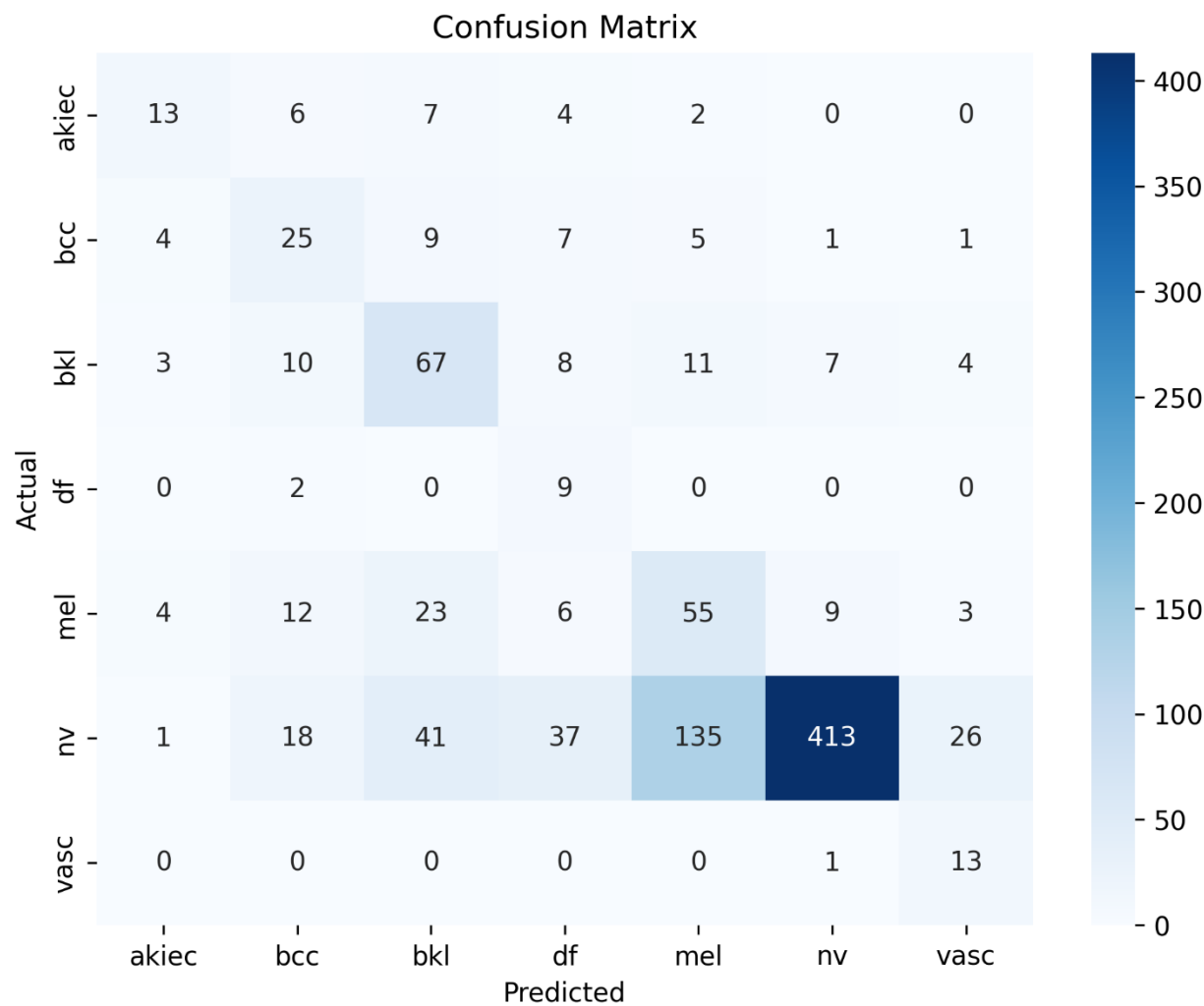


Figure 4.3: Confusion Matrix of the Proposed Model

The confusion matrix indicates that the model correctly classified most images belonging to the **Nevus (nv)** category because this class contains the largest number of training samples.

Some misclassification occurred between **Melanoma (mel)** and **Benign Keratosis (bkl)**, as well as between **Basal Cell Carcinoma (bcc)** and other visually similar lesions. These errors mainly resulted from class imbalance and the high visual similarity among certain lesion categories

4.5 Classification Report Analysis

The classification report summarizes the precision, recall, and F1-score of each skin lesion category.



classification_report.txt

Table 4.2 Classification Report

The evaluation results show that the proposed model achieved the highest precision for the **Nevus (nv)** class. Minority classes such as **Dermatofibroma (df)** and **Vascular Lesions (vasc)** obtained comparatively lower precision due to the limited number of training samples.

The macro-average scores indicate that the model performs reasonably well across multiple classes while still leaving room for improvement through additional data balancing and advanced augmentation techniques.

4.6 Grad-CAM Visualization

To improve model interpretability, Grad-CAM was employed to visualize the regions responsible for the model's predictions.

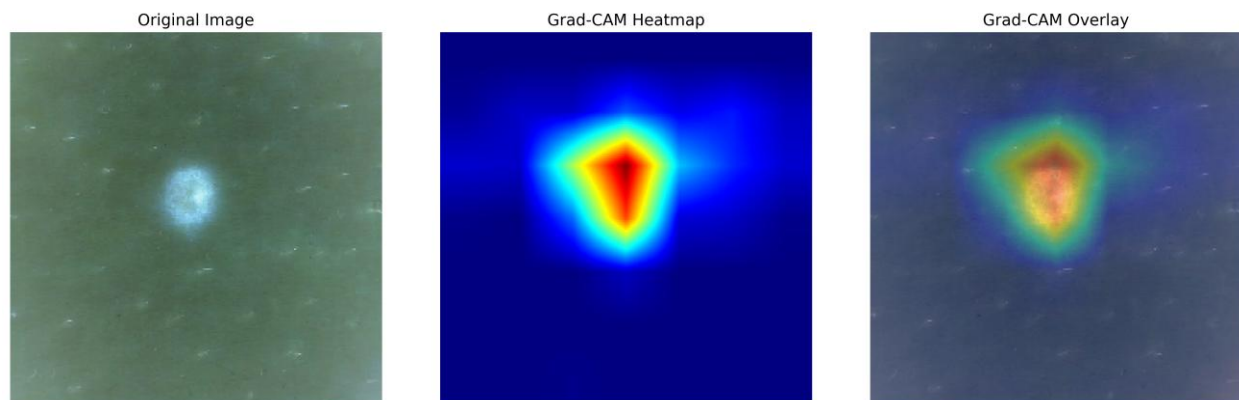


Figure 4.4: Grad-CAM Visualization

The Grad-CAM heatmap clearly highlights the lesion region that influenced the model's prediction. The red and yellow regions indicate areas with higher importance, whereas blue regions represent less significant areas.

The visualization confirms that the proposed model focuses primarily on the lesion instead of irrelevant background regions, thereby increasing the transparency and reliability of the classification process.

4.7 Discussion

The experimental results demonstrate that EfficientNetB0 is capable of learning meaningful features from dermoscopic skin lesion images through transfer learning.

The obtained testing accuracy of **59.38%** indicates that the proposed framework performs reasonably well for multi-class skin cancer classification. The confusion matrix and classification report reveal that majority classes achieve higher prediction accuracy than minority classes, primarily due to dataset imbalance.

The integration of Grad-CAM significantly enhances model interpretability by providing visual explanations of prediction decisions. Such explainability is particularly important in medical applications because clinicians require transparent and trustworthy AI systems before incorporating them into clinical practice.

Overall, the proposed framework successfully combines high-performance image classification with Explainable Artificial Intelligence, making it a promising approach for computer-aided skin cancer diagnosis

4.8 Chapter Summary

This chapter presented the experimental results of the proposed EfficientNetB0-based explainable deep learning framework. The model achieved a testing accuracy of **59.38%** on the HAM10000 dataset. Performance was evaluated using accuracy, confusion matrix, precision, recall, and F1-score. Grad-CAM visualizations further demonstrated that the model successfully identified clinically relevant lesion regions while providing transparent explanations for its predictions.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This research presented an explainable deep learning framework for multi-class skin cancer classification using the HAM10000 dermoscopic image dataset. The proposed framework combined transfer learning with EfficientNetB0 and Explainable Artificial Intelligence (XAI) through Gradient-weighted Class Activation Mapping (Grad-CAM) to improve both classification performance and model interpretability.

A comprehensive preprocessing pipeline was implemented, including image resizing, normalization, data augmentation, and dataset splitting. EfficientNetB0 was selected as the backbone network because of its excellent balance between computational efficiency and classification accuracy. The model was trained using transfer learning and evaluated using standard performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

Experimental results showed that the proposed model achieved a **testing accuracy of 59.38%** on the HAM10000 dataset. Although the overall accuracy indicates moderate classification performance, the model successfully learned discriminative features for most skin lesion categories. The confusion matrix and classification report demonstrated that the model performed particularly well for the majority class (Melanocytic Nevus), while some minority classes remained challenging because of dataset imbalance.

Furthermore, the integration of Grad-CAM significantly enhanced the transparency of the proposed framework by highlighting the image regions responsible for the model's predictions. These visual explanations increase clinician confidence and improve the practical applicability of Artificial Intelligence in medical diagnosis.

Overall, the proposed framework demonstrates that combining EfficientNetB0 with Grad-CAM provides an effective and interpretable solution for automated multi-class skin cancer classification.

5.2 Limitations

Although the proposed framework achieved promising results, several limitations remain.

First, the HAM10000 dataset is highly imbalanced, with certain lesion categories containing significantly fewer samples than others. This imbalance affects the classification performance of minority classes.

Second, only a single public dataset was used for training and evaluation. Testing the model on additional independent datasets would improve its generalization capability.

Third, the proposed framework employed EfficientNetB0 only. More advanced architectures, such as EfficientNetV2, Vision Transformer (ViT), ConvNeXt, or hybrid deep learning models, may provide better classification performance.

Finally, the current study focused only on offline image classification and did not include real-time deployment in a clinical environment.

5.3 Future Work

Several improvements can be considered in future research to enhance the performance and applicability of the proposed framework.

Future work may include:

- Collecting larger and more balanced dermoscopic image datasets.
- Applying advanced data augmentation and class balancing techniques.
- Investigating newer deep learning architectures such as EfficientNetV2, Vision Transformer (ViT), ConvNeXt, and hybrid CNN-transformer models.
- Comparing multiple Explainable Artificial Intelligence techniques, including Grad-CAM++, SHAP, and LIME.
- Developing a web-based or mobile application for real-time skin lesion diagnosis.
- Evaluating the proposed framework using clinical datasets collected from hospitals and dermatology centers.
- Integrating the system into clinical decision support systems to assist dermatologists during diagnosis.

With these improvements, the proposed explainable deep learning framework can become a more reliable and practical tool for early skin cancer detection and medical decision support.

ABSTRACT

Skin cancer is one of the most common and life-threatening diseases worldwide. Early diagnosis plays a crucial role in reducing mortality and improving treatment outcomes. In recent years, deep learning techniques have demonstrated remarkable success in medical image analysis, particularly in automated skin lesion classification. However, most deep learning models operate as black-box systems, making it difficult for clinicians to understand the reasoning behind their predictions.

This research proposes an explainable deep learning framework for multi-class skin cancer classification using the HAM10000 dermoscopic image dataset. EfficientNetB0 was employed as the backbone network through transfer learning because of its high classification performance and computational efficiency. The dataset was preprocessed using image resizing, normalization, and data augmentation techniques to improve model generalization. The model was trained and evaluated using standard performance metrics, including accuracy, precision, recall, F1-score, and confusion matrix.

To improve model interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) was integrated into the proposed framework. Grad-CAM generates visual heatmaps that highlight the regions responsible for the model's predictions, enabling clinicians to better understand the decision-making process.

Experimental results demonstrated that the proposed model achieved a testing accuracy of **59.38%** on the HAM10000 dataset. The Grad-CAM visualizations confirmed that the model primarily focused on lesion regions during classification, thereby increasing transparency and reliability. The proposed framework shows the potential of combining EfficientNetB0 with Explainable Artificial Intelligence for automated and interpretable skin cancer diagnosis.

Keywords: Skin Cancer, Deep Learning, EfficientNetB0, Transfer Learning, Explainable Artificial Intelligence, Grad-CAM, HAM10000

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